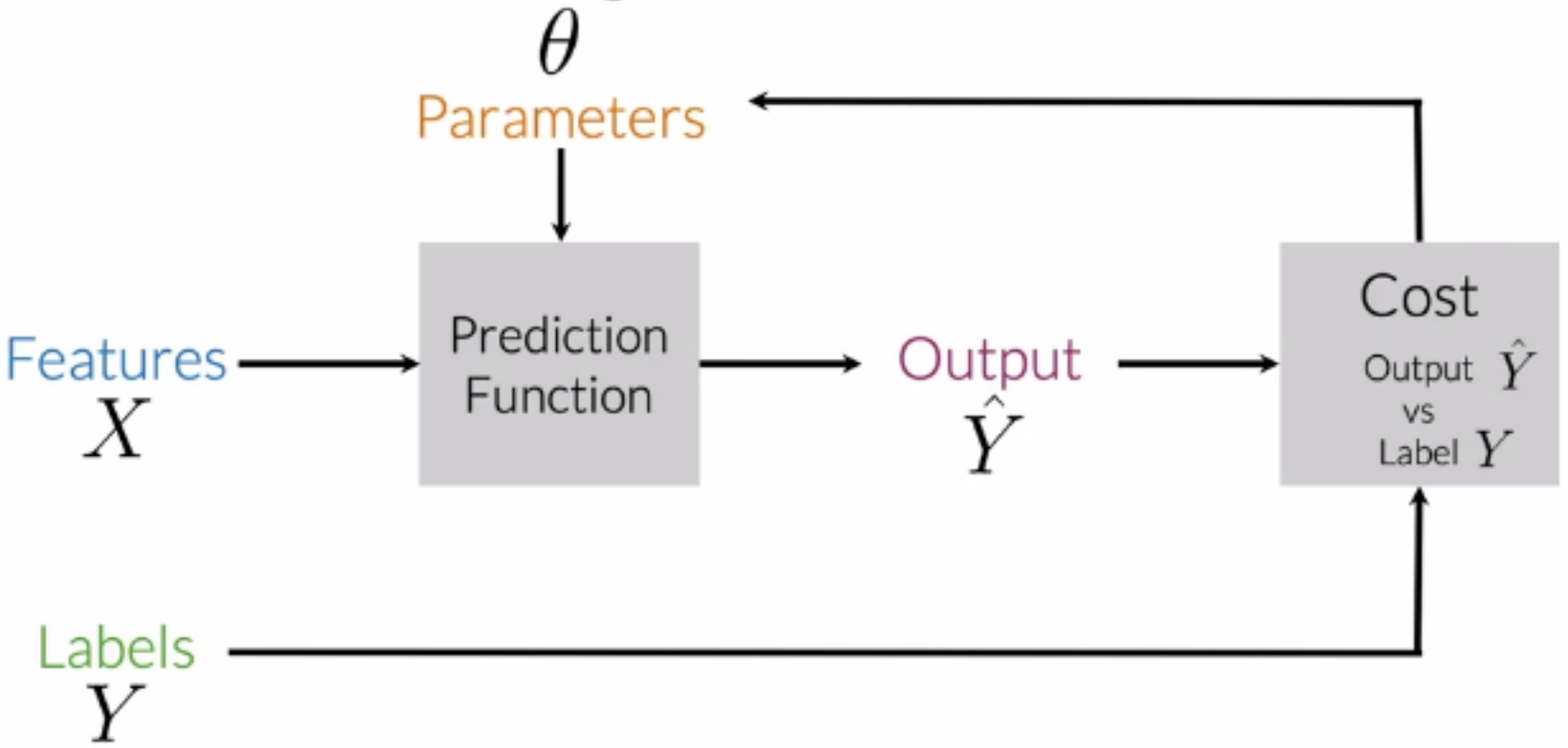
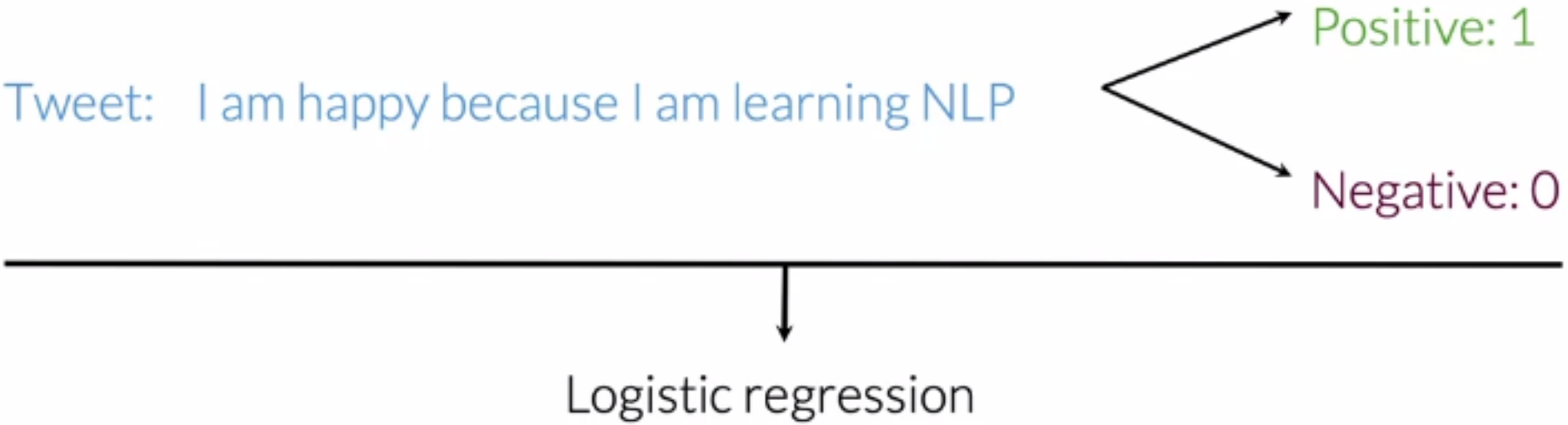
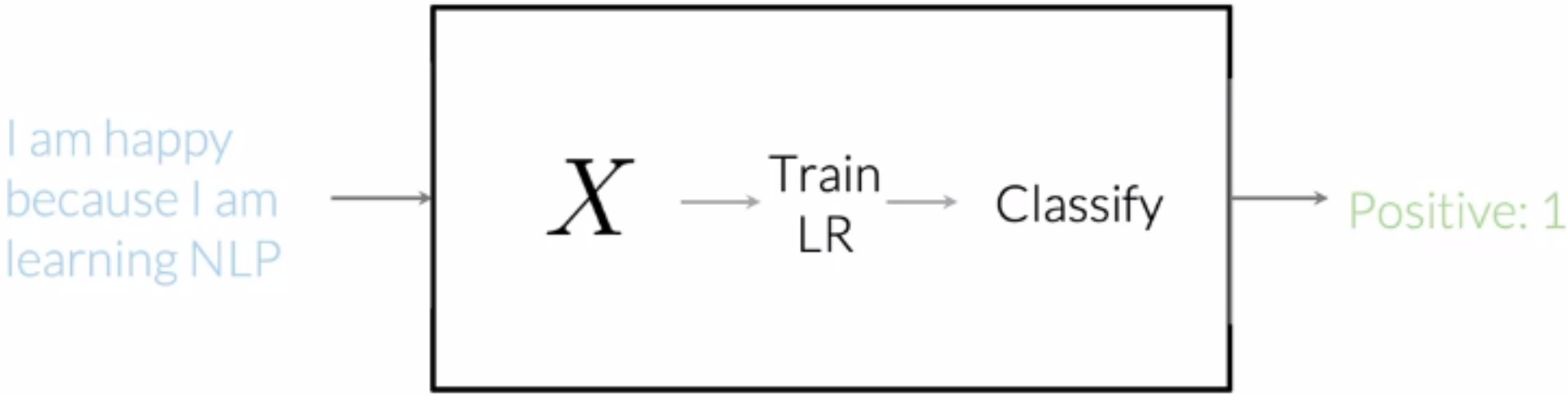
Supervised ML (training)



Sentiment analysis



Sentiment analysis



Vocabulary

Tweets:

[tweet_1, tweet_2, ..., tweet_m]



I am happy because I am learning NLP

...

...

...

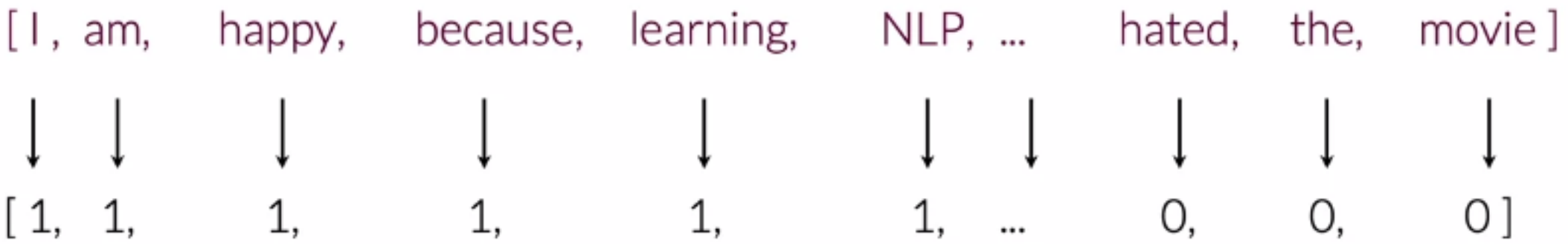
I hated the movie

$$V =$$

[I, am, happy, because, learning, NLP, ... hated, the, movie]

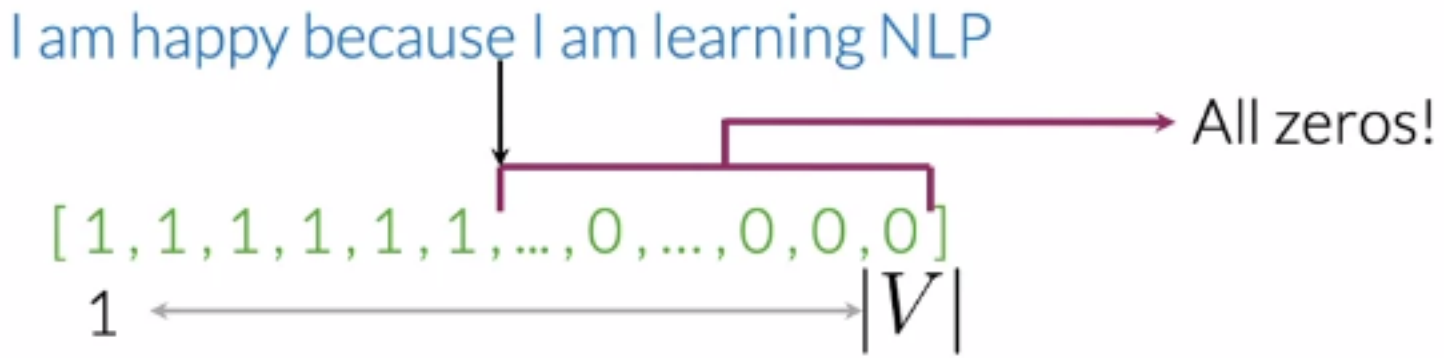
Feature extraction

I am happy because I am learning NLP

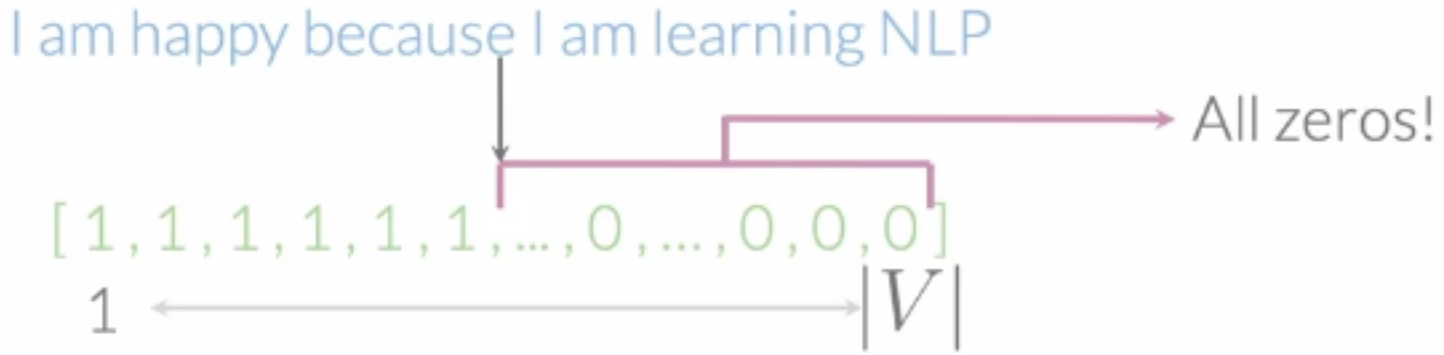


A lot of zeros! That's a sparse representation.

Problems with sparse representations



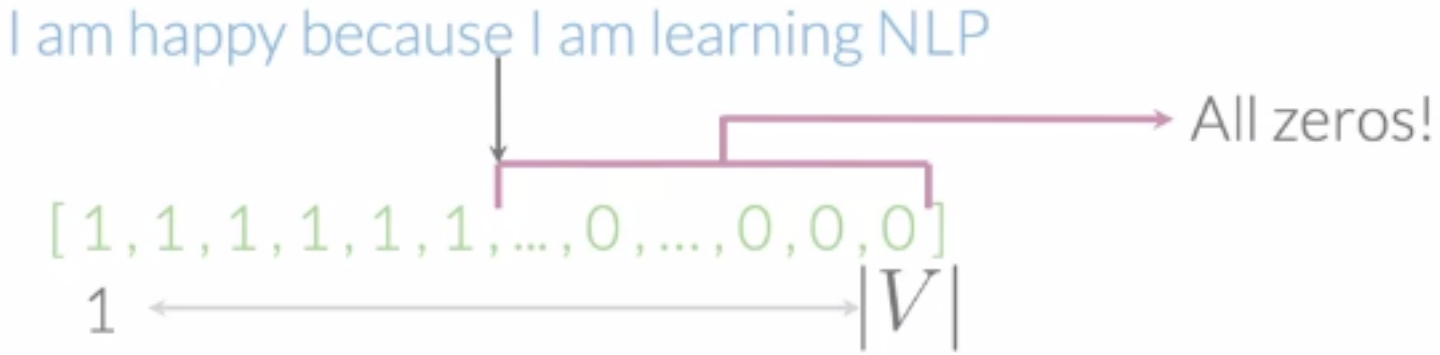
Problems with sparse representations



$$[\theta_0, \theta_1, \theta_2, \ldots, \theta_n]$$

$$n = |V|$$

Problems with sparse representations



$$[\theta_0, \theta_1, \theta_2, \ldots, \theta_n]$$

$$n = |V|$$

1. Large training time
2. Large prediction time

Positive and negative counts

Corpus

I am happy because I am learning NLP

I am happy

I am sad, I am not learning NLP

I am sad

Positive and negative counts

Corpus

I am happy because I am learning NLP

I am happy

I am sad, I am not learning NLP

I am sad

Vocabulary
I
am
happy
because
learning
NLP
sad
not

Positive and negative counts

Positive tweets	Negative tweets
I am happy because I am learning NLP	I am sad, I am not learning NLP
I am happy	I am sad

Positive and negative counts

Positive tweets

I am happy because I am learning NLP
I am happy

Vocabulary
I
am
happy
because
learning
NLP
sad
not

Positive and negative counts

Positive tweets

I am happy because I am learning NLP

I am happy

Vocabulary	PosFreq (1)
I	3
am	3
happy	2
because	1
learning	1
NLP	1
sad	0
not	0

Positive and negative counts

Vocabulary	NegFreq (0)
I	3
am	3
happy	0
because	0
learning	1
NLP	1
sad	2
not	1

Negative tweets

I am sad, I am not learning NLP

I am sad

Word frequency in classes

Vocabulary	PosFreq (1)	NegFreq (0)
I	3	3
am	3	3
happy	2	0
because	1	0
learning	1	1
NLP	1	1
sad	0	1
not	0	1

Word frequency in classes

Vocabulary	PosFreq (1)	NegFreq (0)
I	3	3
am	3	3
happy	2	0
because	1	0
learning	1	1
NLP	1	1
sad	0	1
not	0	1

freqs: dictionary mapping from (word, class) to frequency

Word frequency in classes

Vocabulary	PosFreq (1)	NegFreq (0)
I	3	3
am	3	3
happy	2	0
because	1	0
learning	1	1
NLP	1	1
sad	0	2
not	0	1

freqs: dictionary mapping from (word, class) to frequency

Feature extraction

freqs: dictionary mapping from (word, class) to frequency

Feature extraction

freqs: dictionary mapping from (word, class) to frequency

$$X_m = [\quad]$$



Features of tweet m

Feature extraction

freqs: dictionary mapping from (word, class) to frequency

$$X_m = [1, \quad]$$

Features of tweet m

Bias

Feature extraction

freqs: dictionary mapping from (word, class) to frequency

$$X_m = [1, \sum_w freqs(w, 1), \quad]$$

Features of tweet m

Bias

Sum Pos. Frequencies

Feature extraction

freqs: dictionary mapping from (word, class) to frequency

$$X_m = [1, \sum_w freqs(w, 1), \sum_w freqs(w, 0)]$$

Features of tweet m

Bias

Sum Pos. Frequencies

Sum Neg. Frequencies

Feature extraction

I am sad, I am not learning NLP

Feature extraction

Vocabulary	PosFreq (1)
I	3
am	3
happy	2
because	1
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NLP	1
sad	0
not	0

I am sad, I am not learning NLP

Feature extraction

Vocabulary	PosFreq (1)
I	3
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not	0

I am sad, I am not learning NLP

$$X_m = [1, \sum_w freqs(w, 1), \sum_w freqs(w, 0)]$$

Feature extraction

Vocabulary	PosFreq (1)
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not	0

I am sad, I am not learning NLP

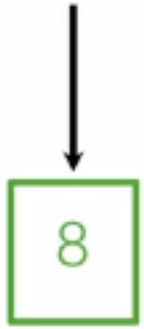
$$X_m = [1, \sum_w freqs(w, 1), \sum_w freqs(w, 0)]$$

Feature extraction

Vocabulary	PosFreq (1)
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I am sad, I am not learning NLP

$$X_m = [1, \sum_w freqs(w, 1), \sum_w freqs(w, 0)]$$



Feature extraction

Vocabulary	NegFreq (0)
I	3
am	3
happy	0
because	0
learning	1
NLP	1
sad	2
not	1

I am sad, I am not learning NLP

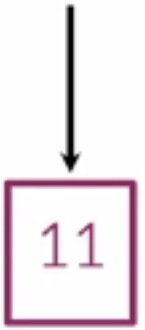
$$X_m = [1, \sum_w freqs(w, 1), \sum_w freqs(w, 0)]$$

Feature extraction

Vocabulary	NegFreq (0)
I	3
am	3
happy	0
because	0
learning	1
NLP	1
sad	2
not	1

I am sad, I am not learning NLP

$$X_m = [1, \sum_w freqs(w, 1), \sum_w freqs(w, 0)]$$



Feature extraction

I am sad, I am not learning NLP

$$X_m = [1, \sum_w freqs(w, 1), \sum_w freqs(w, 0)]$$

Feature extraction

I am sad, I am not learning NLP

$$X_m = [1, \sum_w freqs(w, 1), \sum_w freqs(w, 0)]$$

$$\downarrow$$

$$X_m = [1, 8, 11]$$

Preprocessing: stop words and punctuation

@YMourri and @AndrewYNg are
tuning a GREAT AI model at
https://deeplearning.ai!!!

Preprocessing: stop words and punctuation

@YMourri and @AndrewYNg are tuning a GREAT AI model at https://deeplearning.ai!!!

Stop words
and
is
are
at
has
for
a

Punctuation
,
.
:
!
“
‘

Preprocessing: stop words and punctuation

@YMourri ~~and~~ @AndrewYNg ~~are~~ tuning ~~a~~ GREAT AI model ~~at~~ https://deeplearning.ai!!!

Stop words	Punctuation
and	,
is	.
are	:
at	!
has	"
for	'
a	

Preprocessing: stop words and punctuation

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Stop words
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Preprocessing: stop words and punctuation

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Stop words
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Preprocessing: stop words and punctuation

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@YMourri @AndrewYNg tuning
GREAT AI model
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Stop words
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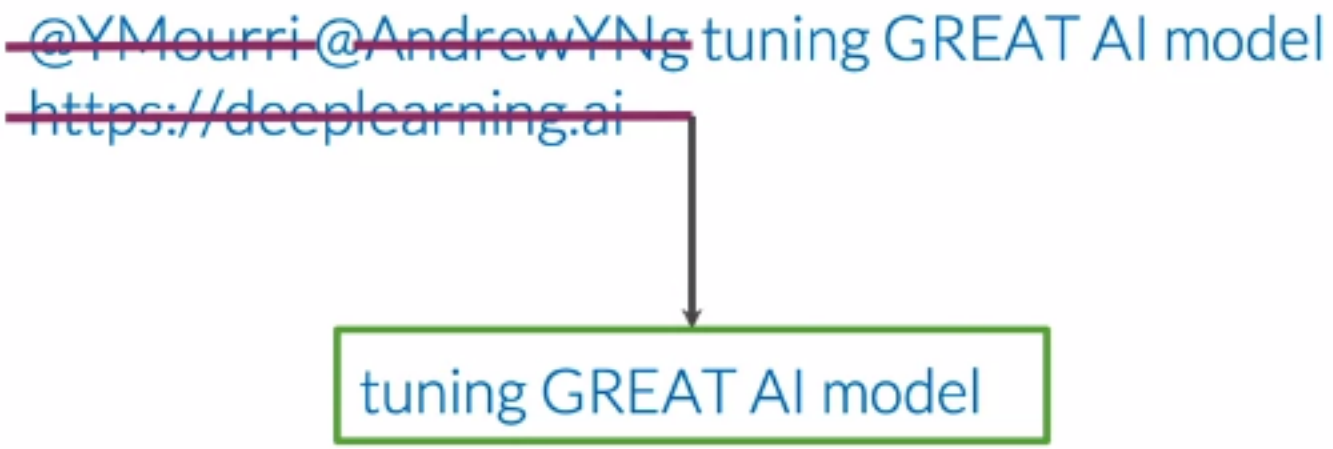
Preprocessing: Handles and URLs

@YMourri @AndrewYNg tuning GREAT AI model
https://deeplearning.ai

Preprocessing: Handles and URLs

~~@YMourri @AndrewYNg~~ tuning GREAT AI model
~~https://deeplearning.ai~~

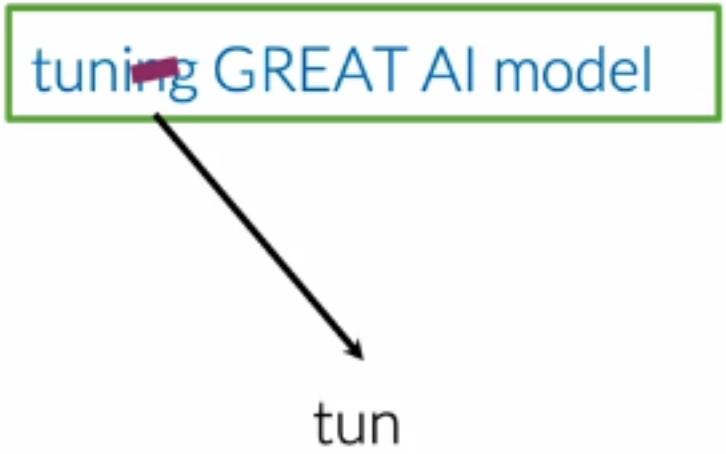
Preprocessing: Handles and URLs



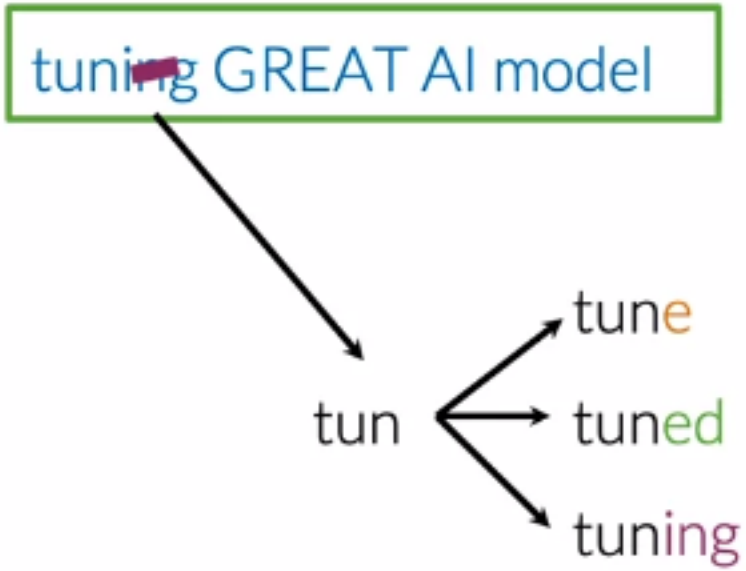
Preprocessing: Stemming and lowercasing

tuning GREAT AI model

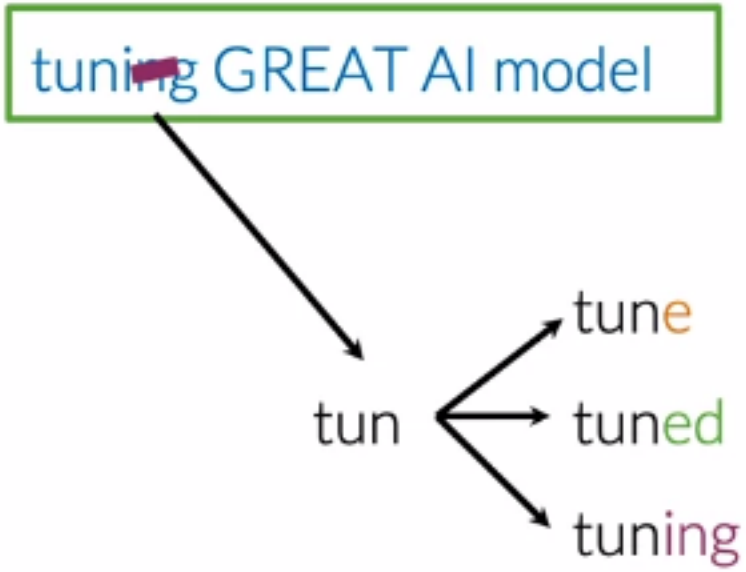
Preprocessing: Stemming and lowercasing



Preprocessing: Stemming and lowercasing



Preprocessing: Stemming and lowercasing

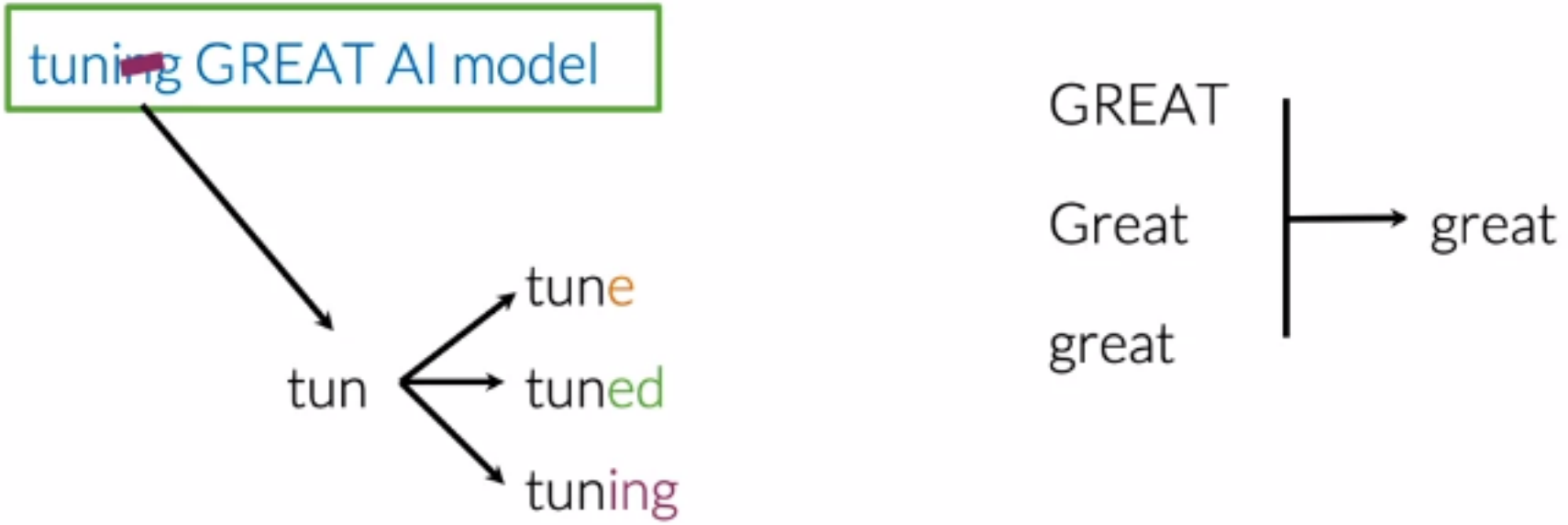


GREAT

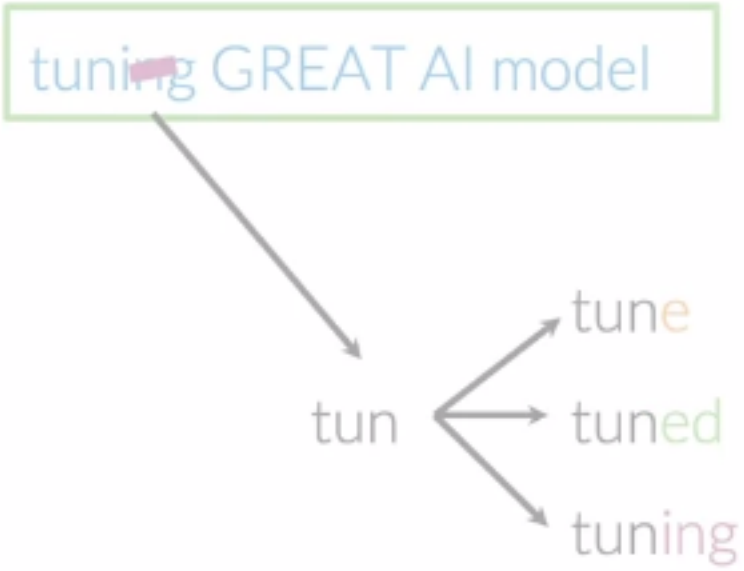
Great

great

Preprocessing: Stemming and lowercasing



Preprocessing: Stemming and lowercasing





Preprocessed tweet:
[tun, great, ai, model]

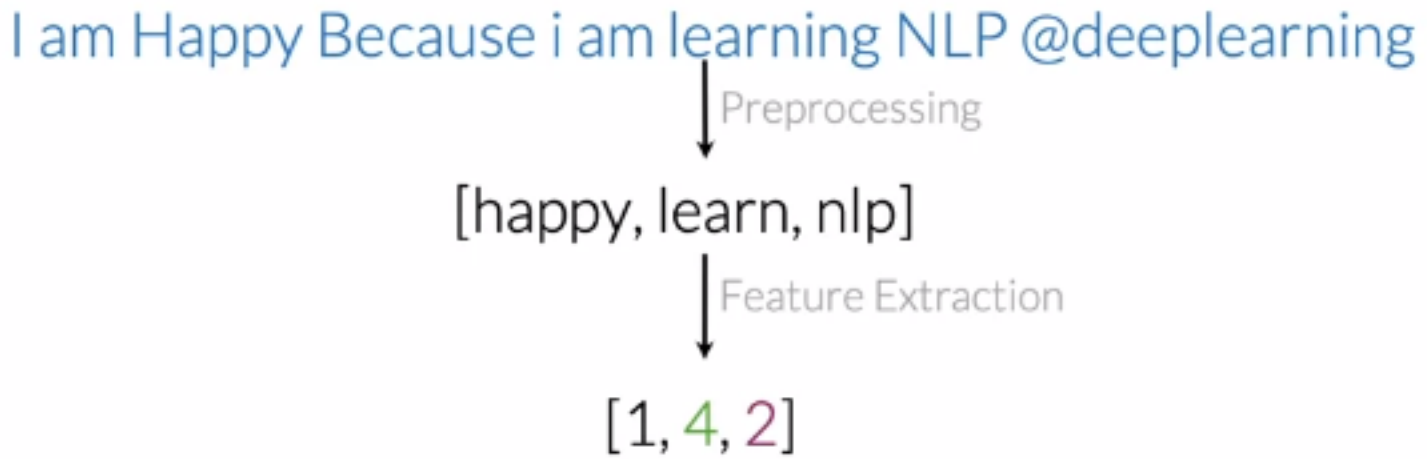
General overview

I am Happy Because i am learning NLP @deeplearning

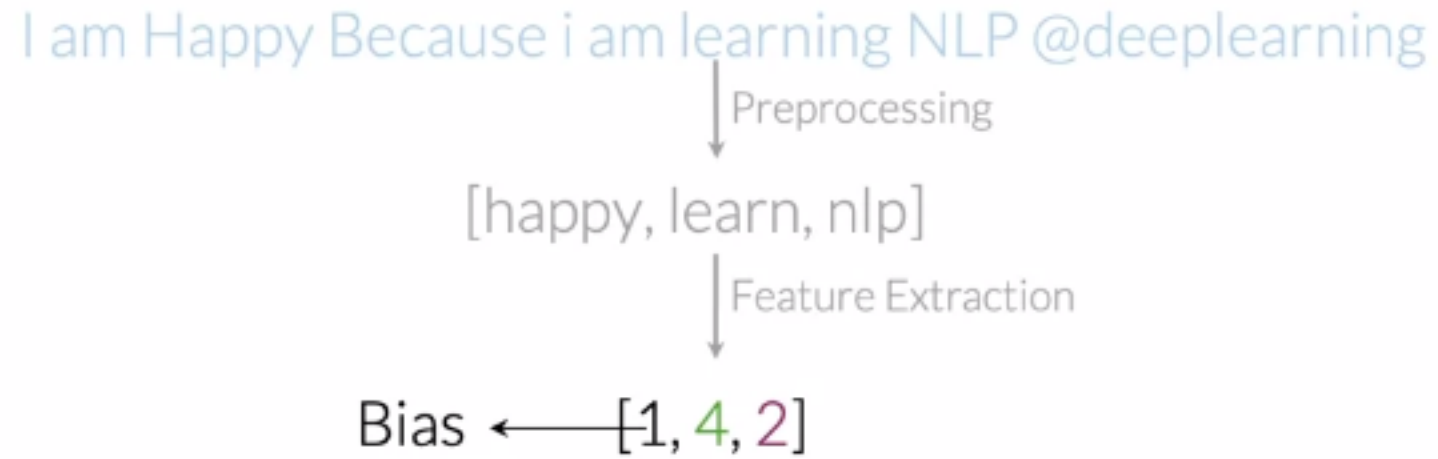


[happy, learn, nlp]

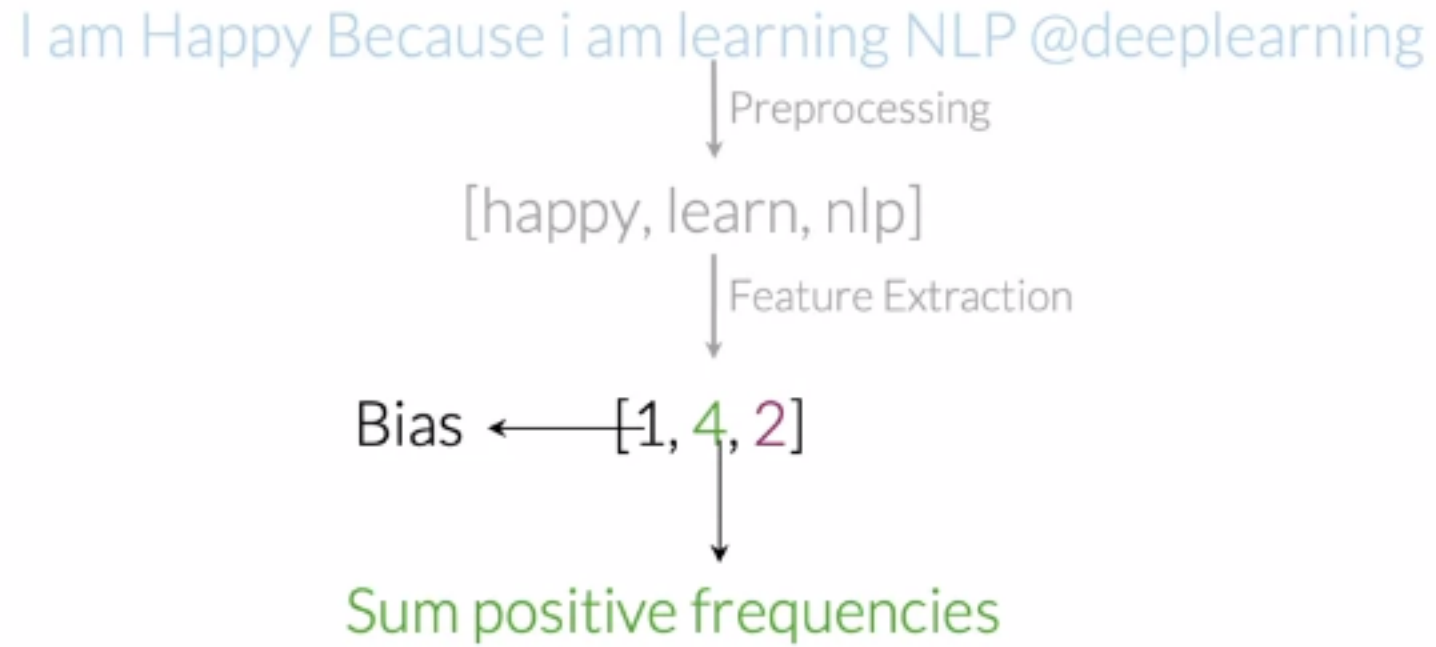
General overview



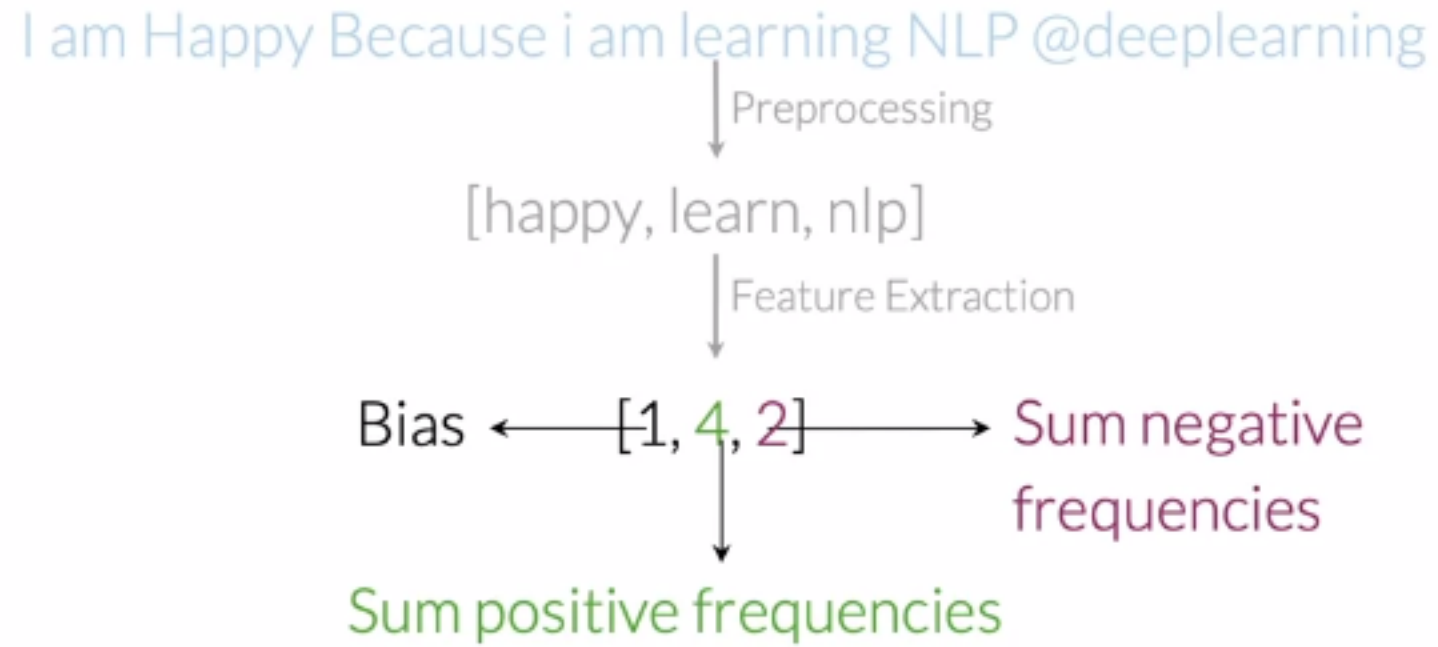
General overview



General overview



General overview



General overview

I am Happy Because i am
learning NLP
@deeplearning

I am sad not learning NLP

...

I am sad :(

General overview

I am Happy Because i am learning NLP @deeplearning

I am sad not learning NLP

...

I am sad :(

[happy, learn, nlp]

[sad, not, learn, nlp]

...

[sad]

General overview

I am Happy Because i am learning NLP @deeplearning

I am sad not learning NLP

...

I am sad :(

[happy, learn, nlp]

[sad, not, learn, nlp]

...

[sad]



[[1, 40, 20],
[1, 20, 50],
...
[1, 5, 35]]

General overview

$$\begin{bmatrix} 1 & X_1^{(1)} & X_2^{(1)} \\ 1 & X_1^{(2)} & X_2^{(2)} \\ \vdots & \vdots & \vdots \\ 1 & X_1^{(m)} & X_2^{(m)} \end{bmatrix} \longleftrightarrow$$

[[1, 40, 20],
[1, 20, 50],
...
[1, 5, 35]]

General Implementation

```
freqs = build_freqs(tweets,labels) #Build frequencies dictionary

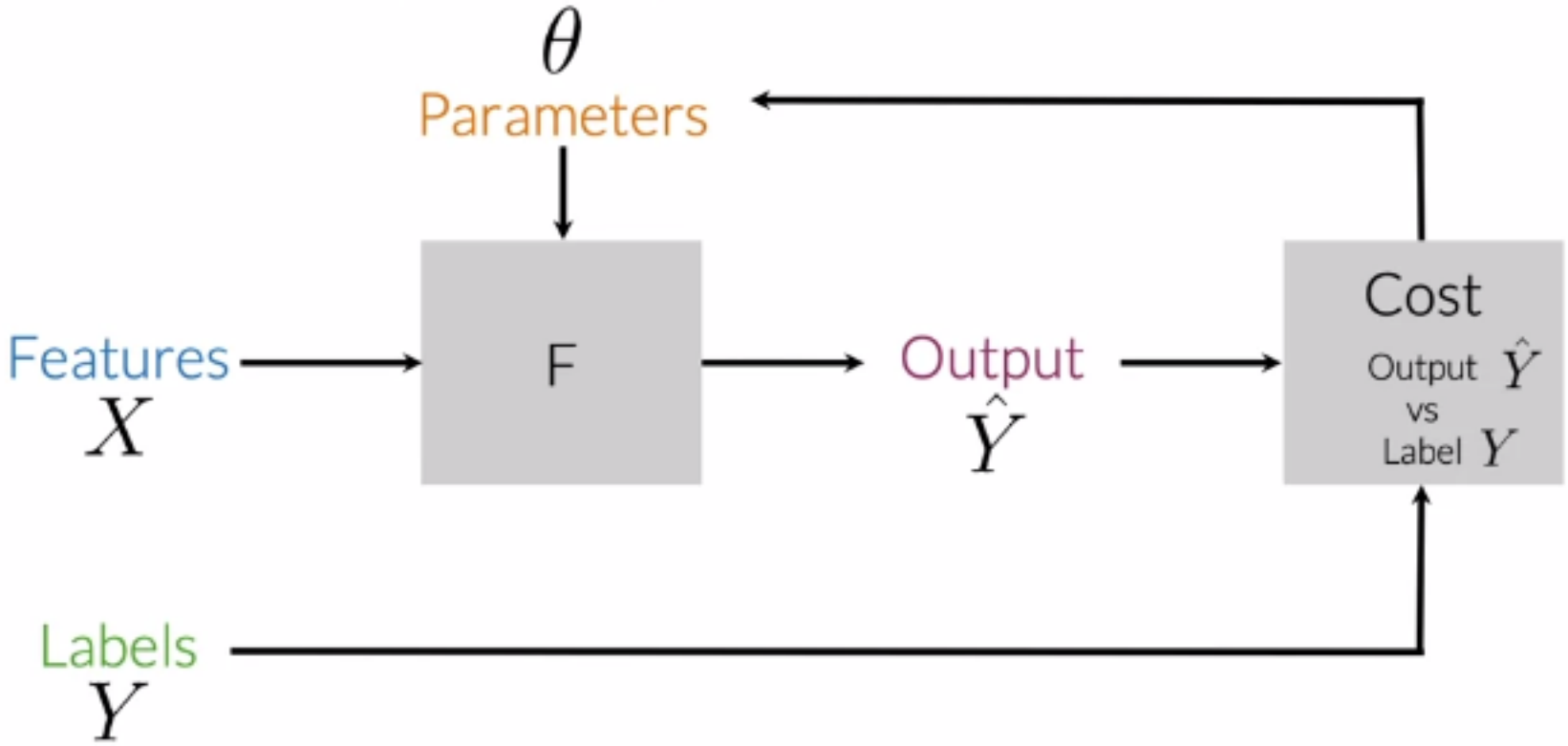
X = np.zeros((m,3)) #Initialize matrix X

for i in range(m): #For every tweet

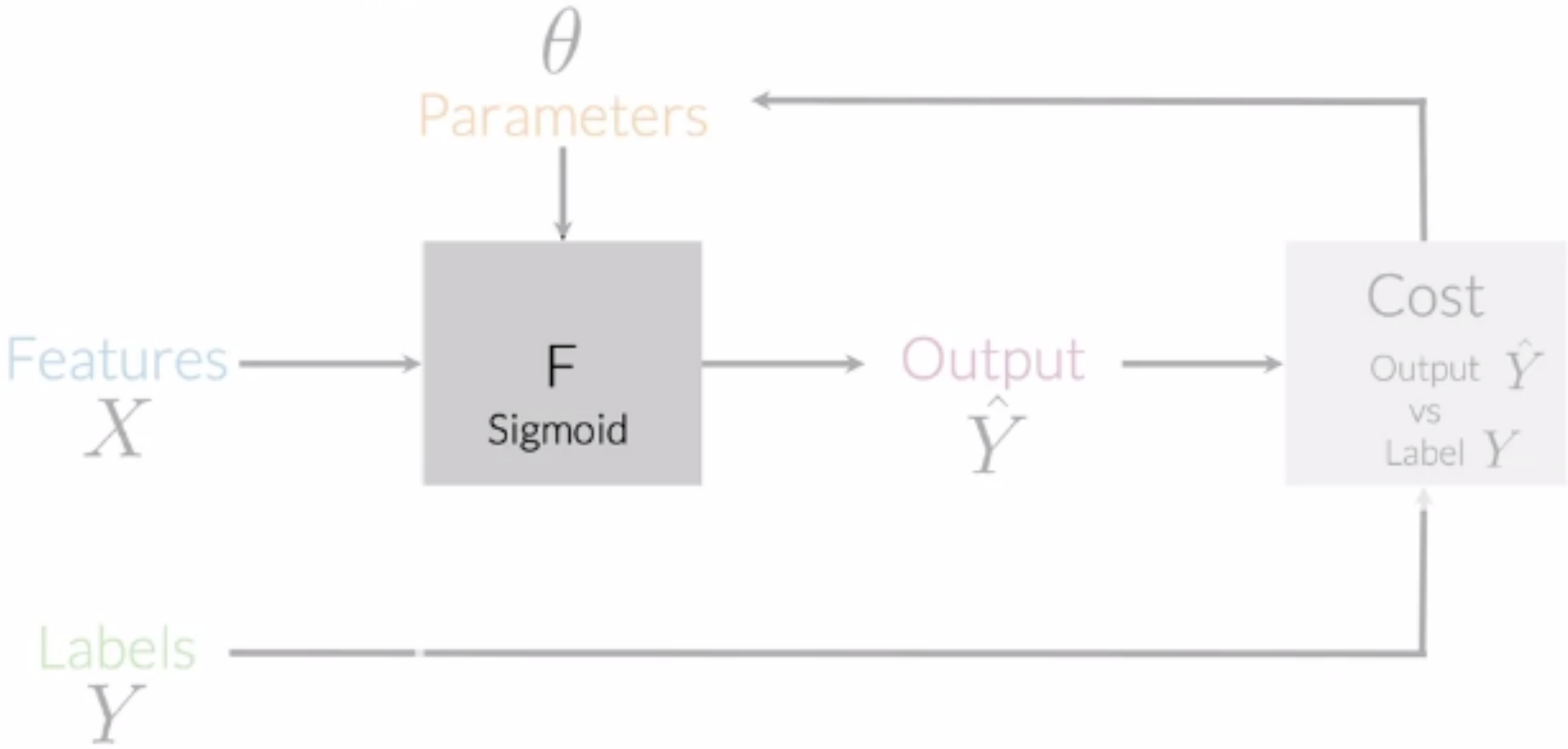
        p_tweet = process_tweet(tweets[i]) #Process tweet

    X[i,:] = extract_features(p_tweet,freqs) #Extract Features
```

Overview of logistic regression



Overview of logistic regression

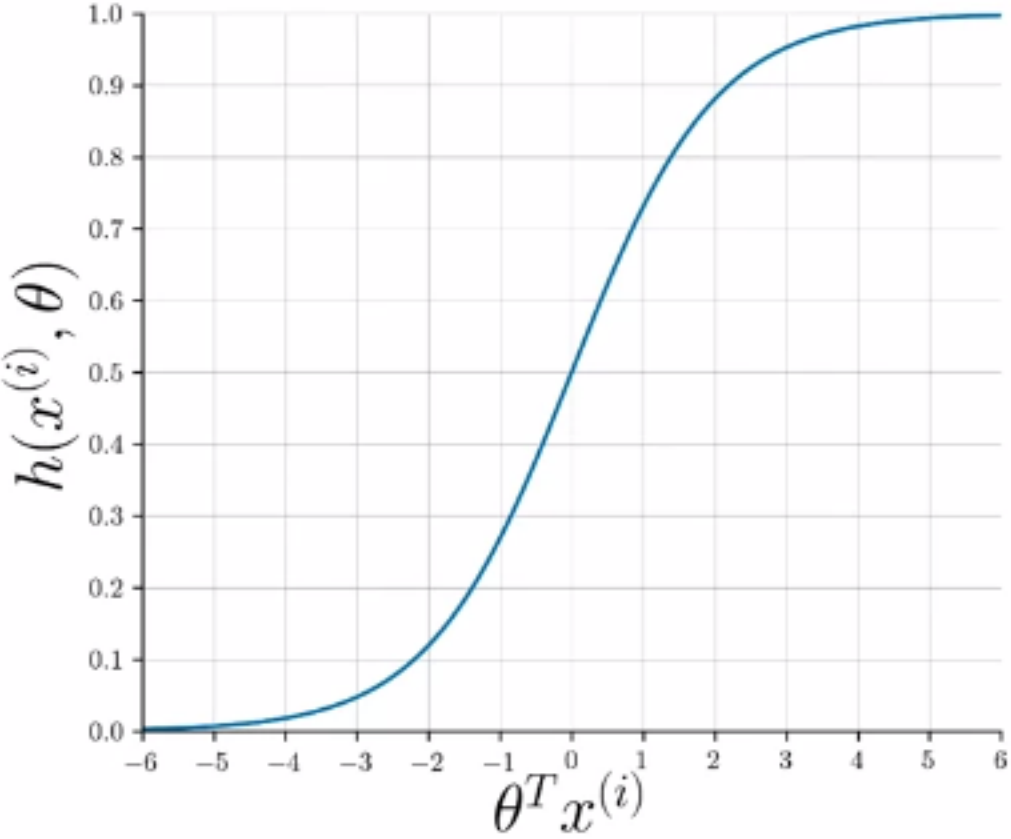


Overview of logistic regression

$$h(x^{(i)}, \theta) = \frac{1}{1 + e^{-\theta^T x^{(i)}}}$$

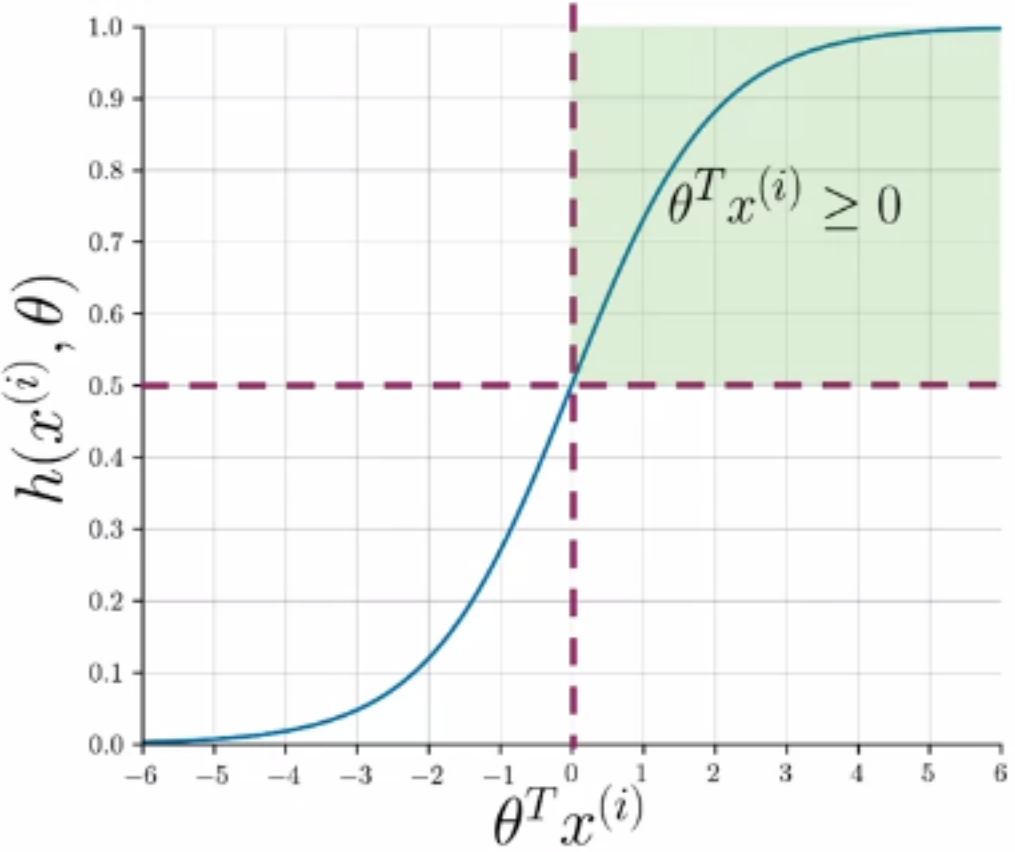
Overview of logistic regression

$$h(x^{(i)}, \theta) = \frac{1}{1 + e^{-\theta^T x^{(i)}}}$$



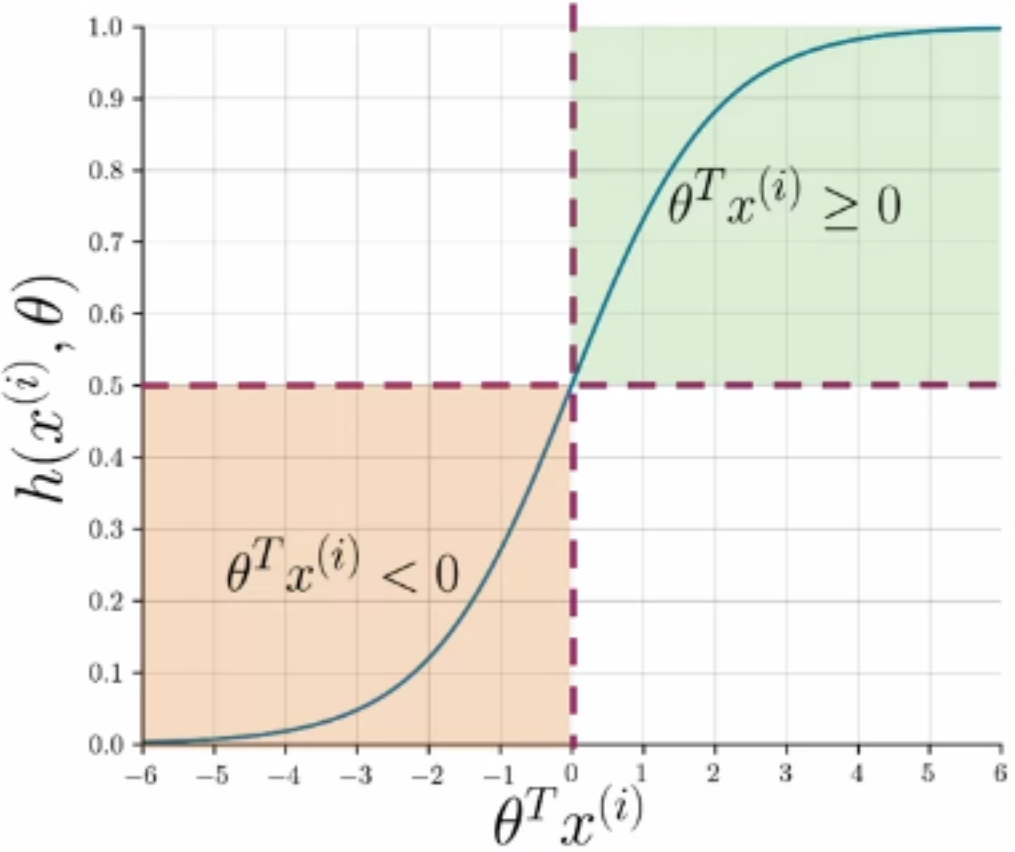
Overview of logistic regression

$$h(x^{(i)}, \theta) = \frac{1}{1 + e^{-\theta^T x^{(i)}}}$$



Overview of logistic regression

$$h(x^{(i)}, \theta) = \frac{1}{1 + e^{-\theta^T x^{(i)}}}$$



Overview of logistic regression

@YMourri and
@AndrewYNg are tuning a
GREAT AI model

Overview of logistic regression

@YMourri and
@AndrewYNg are tuning a
GREAT AI model



[tun, ai, great, model]

Overview of logistic regression

@YMourri and
@AndrewYNg are tuning a
GREAT AI model

[tun, ai, great, model]

$$x^{(i)} = \begin{bmatrix} 1 \\ 3476 \\ 245 \end{bmatrix}$$

Overview of logistic regression

@YMourri and
@AndrewYNg are tuning a
GREAT AI model

↓

[tun, ai, great, model]

↓

$$x^{(i)} = \begin{bmatrix} 1 \\ 3476 \\ 245 \end{bmatrix} \quad \theta = \begin{bmatrix} 0.00003 \\ 0.00150 \\ -0.00120 \end{bmatrix}$$

Overview of logistic regression

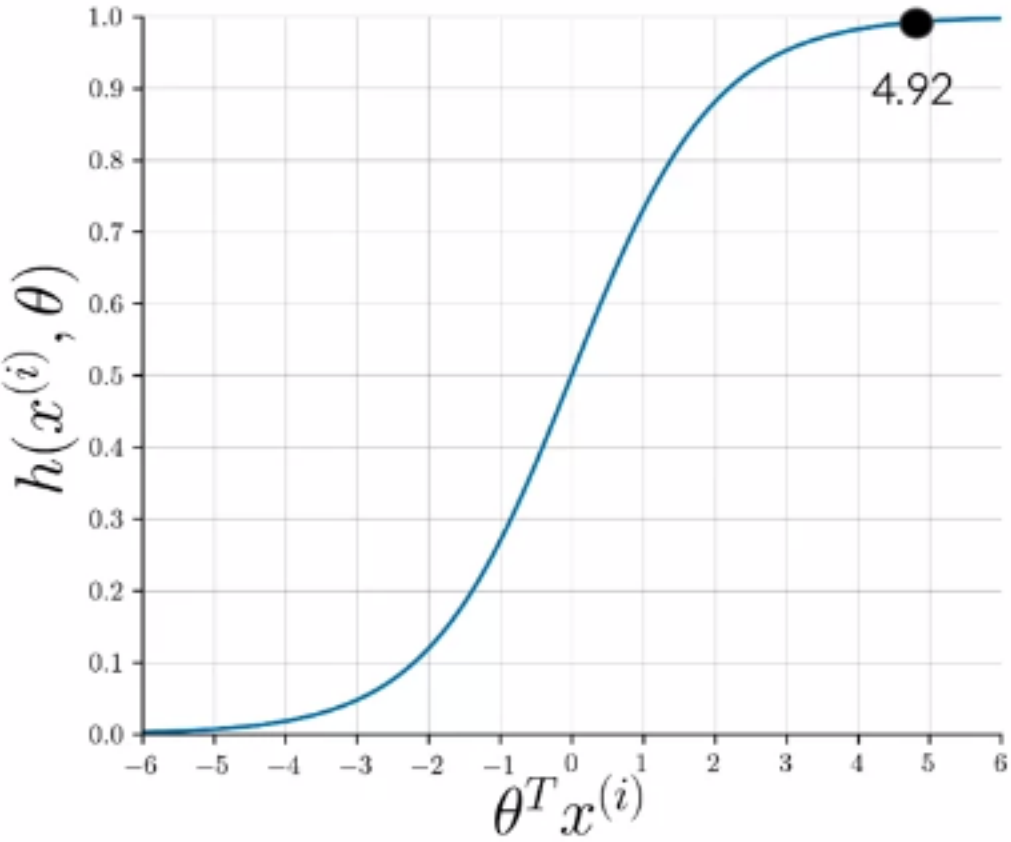
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↓

[tun, ai, great, model]

↓

$$x^{(i)} = \begin{bmatrix} 1 \\ 3476 \\ 245 \end{bmatrix} \quad \theta = \begin{bmatrix} 0.00003 \\ 0.00150 \\ -0.00120 \end{bmatrix}$$



Overview of logistic regression

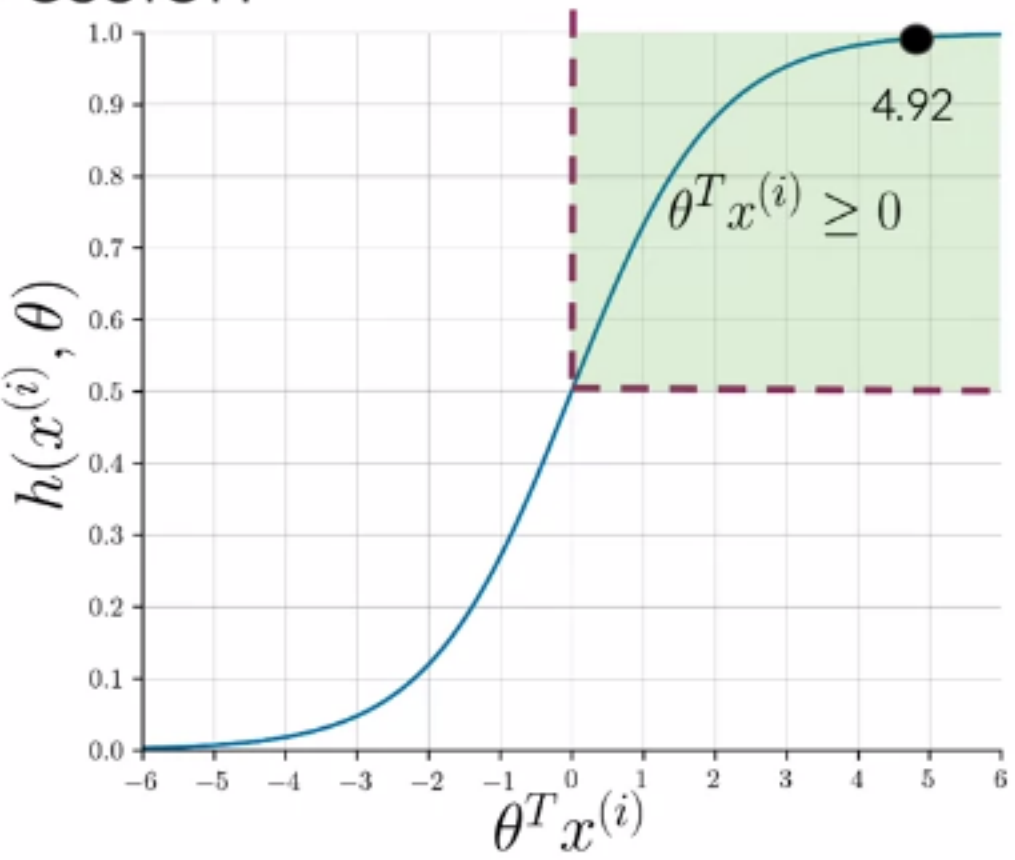
@YMourri and
@AndrewYNg are tuning a
GREAT AI model

↓

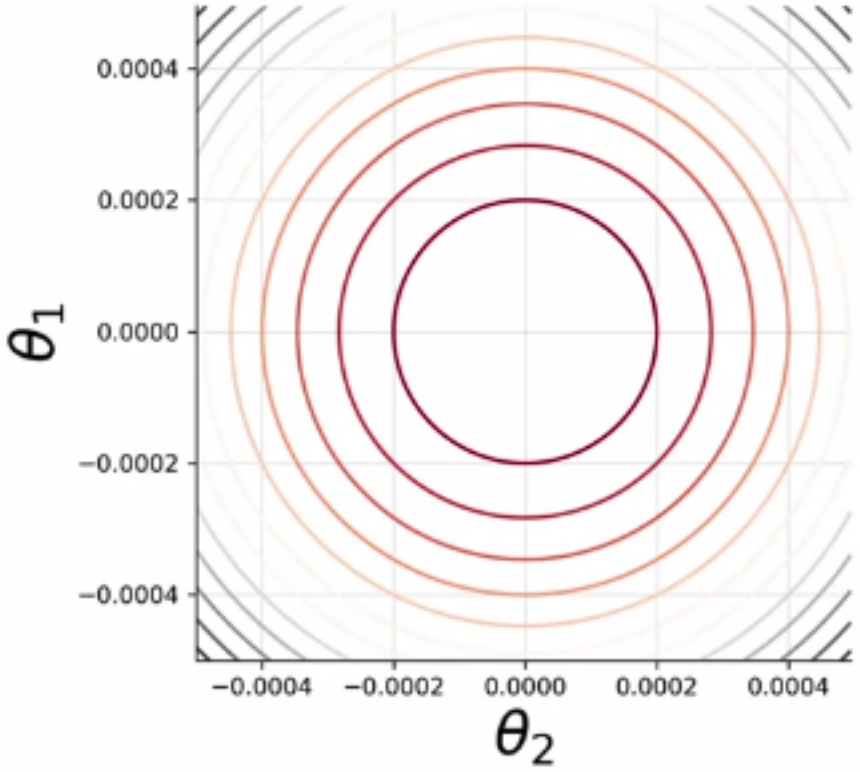
[tun, ai, great, model]

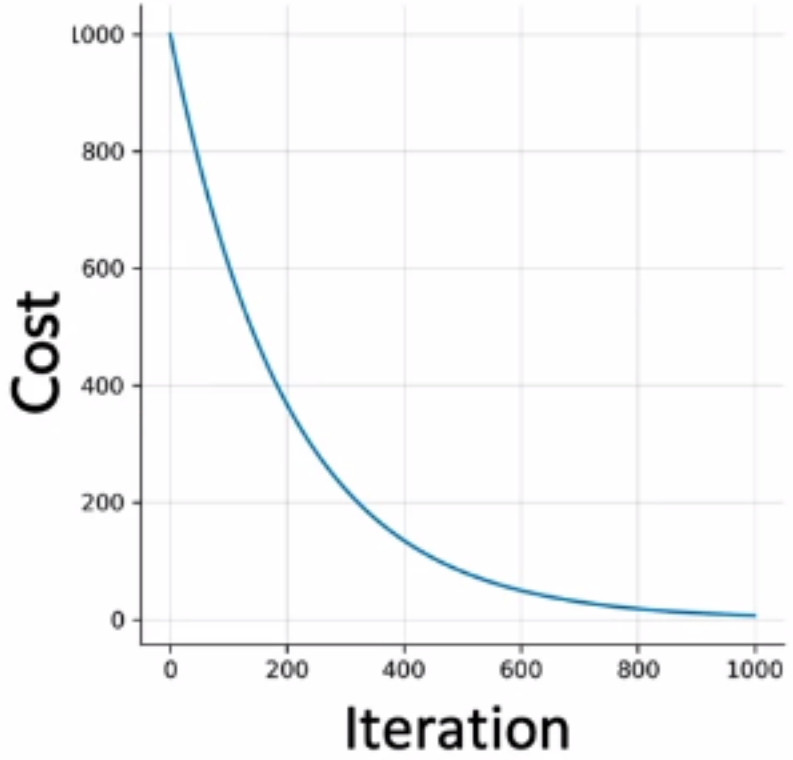
↓

$$x^{(i)} = \begin{bmatrix} 1 \\ 3476 \\ 245 \end{bmatrix} \quad \theta = \begin{bmatrix} 0.00003 \\ 0.00150 \\ -0.00120 \end{bmatrix}$$

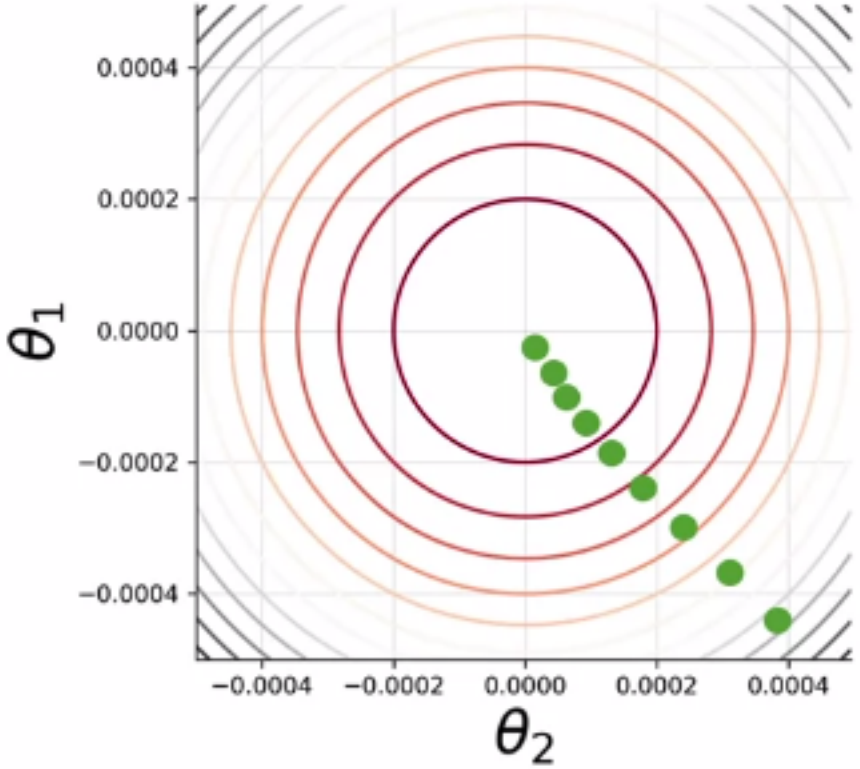


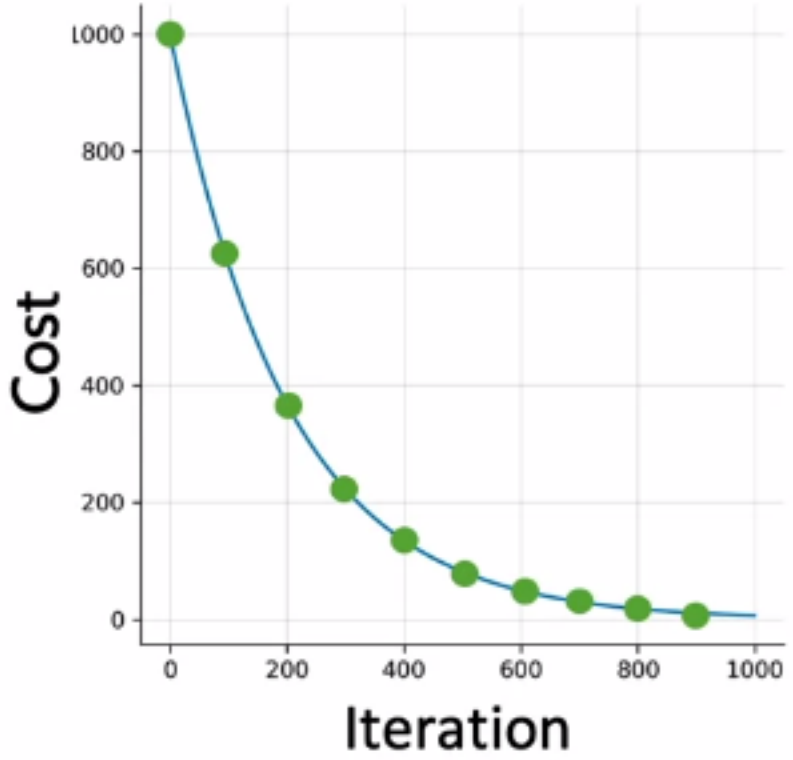
Training LR


0.0004
0.0002
0.0000
−0.0002
−0.0004
θ_1
−0.0004
−0.0002
0.0000
0.0002
0.0004
θ_2


1000
800
600
400
200
0
Cost
0
200
400
600
800
1000
Iteration

Training LR


0.0004
0.0002
0.0000
−0.0002
−0.0004
θ_1
−0.0004
−0.0002
0.0000
0.0002
0.0004
θ_2

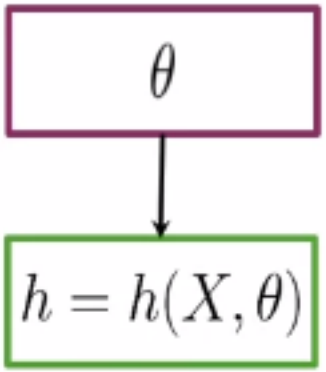

1000
800
600
400
200
0
Cost
0
200
400
600
800
1000
Iteration

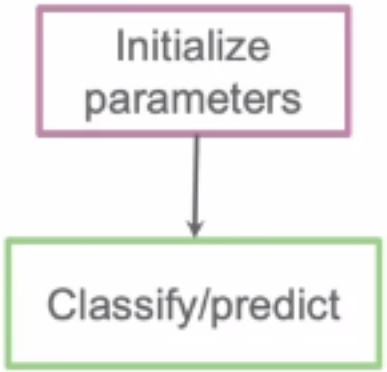
Training LR

θ

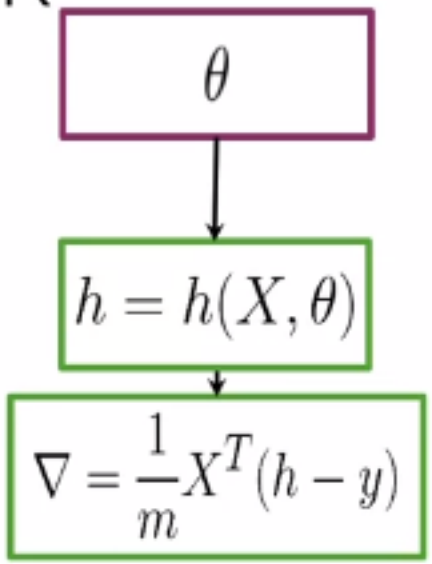
Initialize parameters

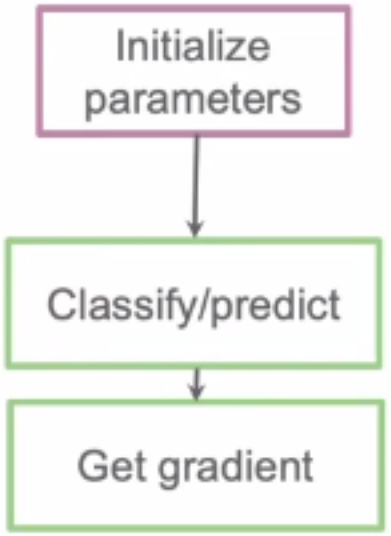
Training LR



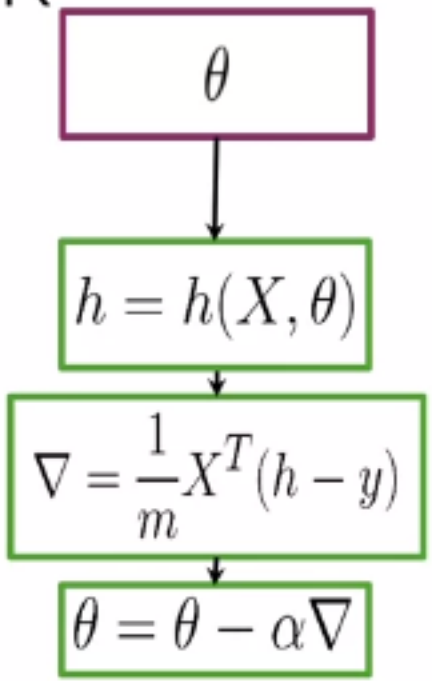


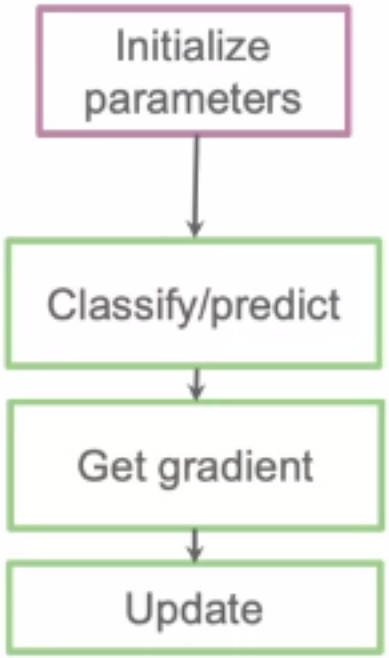
Training LR


θ
$h = h(X, \theta)$
$\nabla = \frac{1}{m} X^T (h - y)$

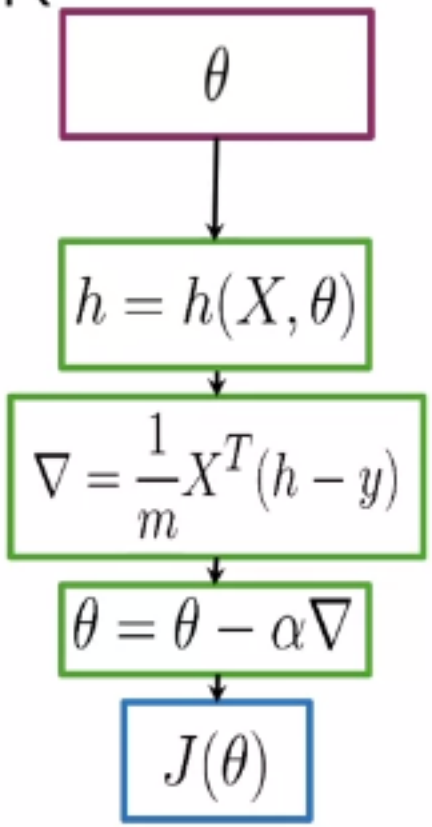

Initialize parameters
Classify/predict
Get gradient

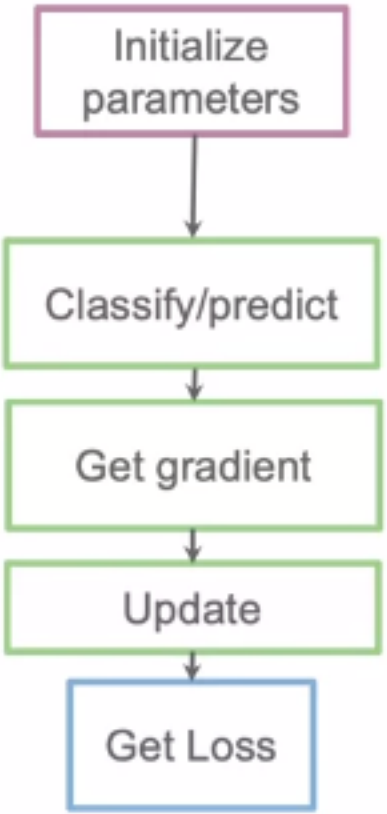
Training LR


θ
$h = h(X, \theta)$
$\nabla = \frac{1}{m} X^T (h - y)$
$\theta = \theta - \alpha \nabla$

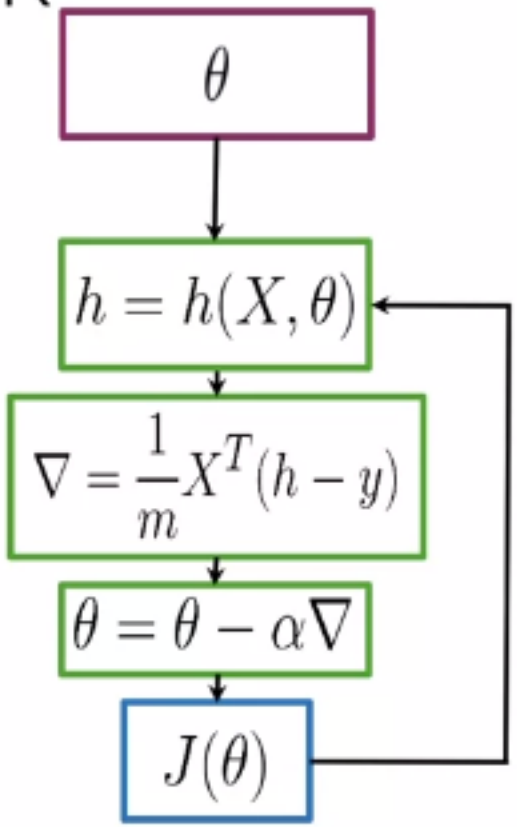

Initialize parameters
Classify/predict
Get gradient
Update

Training LR

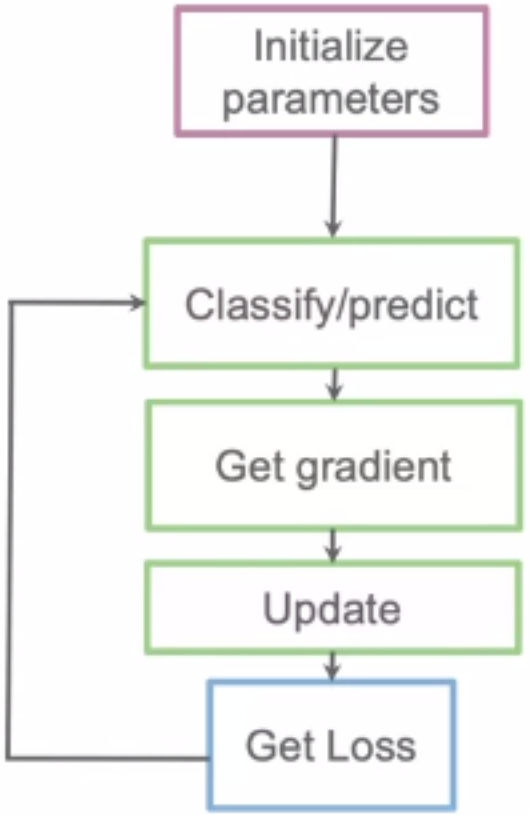

θ
$h = h(X, \theta)$
$\nabla = \frac{1}{m} X^T (h - y)$
$\theta = \theta - \alpha \nabla$
$J(\theta)$


Initialize parameters
Classify/predict
Get gradient
Update
Get Loss

Training LR



Until good enough



Testing logistic regression

- X_{val} Y_{val} θ

Testing logistic regression

- X_{val} Y_{val} θ

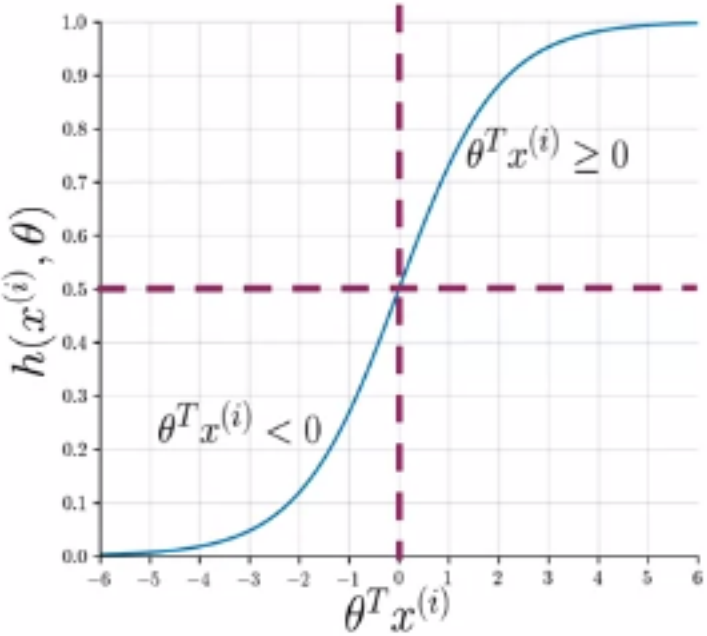
$$h(X_{val}, \theta)$$

Testing logistic regression

- X_{val} Y_{val} θ

  $h(X_{val}, \theta)$

$$pred = h(X_{val}, \theta) \geq 0.5$$



Testing logistic regression

- X_{val} Y_{val} θ

$h(X_{val}, \theta)$

$pred = h(X_{val}, \theta) \geq 0.5$

$$\begin{bmatrix} 0.3 \\ 0.8 \\ 0.5 \\ \vdots \\ h_m \end{bmatrix}$$

Testing logistic regression

- X_{val} Y_{val} θ

$h(X_{val}, \theta)$

$pred = h(X_{val}, \theta) \geq 0.5$

$$\begin{bmatrix} 0.3 \\ 0.8 \\ 0.5 \\ \vdots \\ h_m \end{bmatrix} \geq 0.5$$

Testing logistic regression

- X_{val} Y_{val} θ

$$h(X_{val}, \theta)$$

$$pred = h(X_{val}, \theta) \geq 0.5$$

$$\begin{bmatrix} 0.3 \\ 0.8 \\ 0.5 \\ \vdots \\ h_m \end{bmatrix} \geq 0.5 = \begin{bmatrix} 0.3 \geq 0.5 \\ 0.8 \geq 0.5 \\ 0.5 \geq 0.5 \\ \vdots \\ pred_m \geq 0.5 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ \vdots \\ pred_m \end{bmatrix}$$

Testing logistic regression

- X_{val} Y_{val} θ

$h(X_{val}, \theta)$

$pred = h(X_{val}, \theta) \geq 0.5$

$$\sum_{i=1}^{m} \frac{(pred^{(i)} == y_{val}^{(i)})}{m}$$

Testing logistic regression

- X_{val} Y_{val} θ

$$h(X_{val}, \theta)$$

$$pred = h(X_{val}, \theta) \geq 0.5$$

$$\sum_{i=1}^{m} \frac{(pred^{(i)} == y_{val}^{(i)})}{m}$$

$$\begin{bmatrix} 0 \\ 1 \\ 1 \\ \vdots \\ pred_m \end{bmatrix} == \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ Y_{val_m} \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \\ \vdots \\ pred_m == Y_{val_m} \end{bmatrix}$$

Testing logistic regression

- X_{val} Y_{val} θ

$h(X_{val}, \theta)$

$pred = h(X_{val}, \theta) \geq 0.5$

$$\sum_{i=1}^{m} \frac{(pred^{(i)} == y_{val}^{(i)})}{m}$$

Testing logistic regression

$$Y_{val} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \quad pred = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Testing logistic regression

$$Y_{val} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \quad pred = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (Y_{val} == pred) = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

Testing logistic regression

$$Y_{val} = \begin{bmatrix} 0 \\ 1 \\ \underline{1} \\ 0 \\ 1 \end{bmatrix} \quad pred = \begin{bmatrix} 0 \\ 1 \\ \underline{0} \\ 0 \\ 1 \end{bmatrix} \quad (Y_{val} == pred) = \begin{bmatrix} 1 \\ 1 \\ \underline{0} \\ 1 \\ 1 \end{bmatrix}$$

$$\text{accuracy} = \frac{4}{5} = 0.8$$

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))]$$

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))]$$

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m} \left[y^{(i)} \log h(x^{(i)}, \theta) + (1 - y^{(i)}) \log(1 - h(x^{(i)}, \theta)) \right]$$

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))]$$

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))\right]$$

$y^{(i)}$ $h(x^{(i)},\theta)$

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))\right]$$

$y^{(i)}$	$h(x^{(i)},\theta)$	
0	any	0

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))\right]$$

$y^{(i)}$	$h(x^{(i)},\theta)$	
0	any	0
1	0.99	~0

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))\right]$$

$y^{(i)}$	$h(x^{(i)},\theta)$	
0	any	0
1	0.99	~0
1	~0	-inf

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))]$$

$y^{(i)}$ $h(x^{(i)},\theta)$

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))]$$

$y^{(i)}$	$h(x^{(i)},\theta)$	
1	any	0

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))]$$

$y^{(i)}$	$h(x^{(i)},\theta)$	
1	any	0
0	0.01	~0

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))]$$

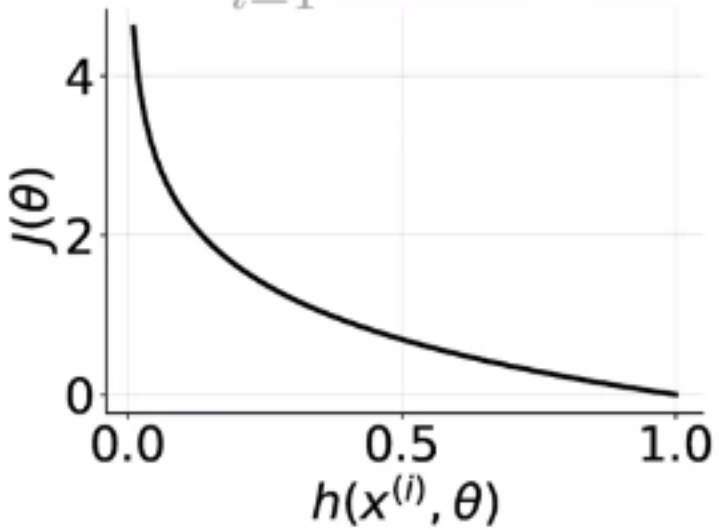
$y^{(i)}$	$h(x^{(i)},\theta)$	
1	any	0
0	0.01	~0
0	~1	-Inf

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))\right]$$

Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))]$$



Cost function for logistic regression

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log h(x^{(i)},\theta) + (1-y^{(i)})\log(1-h(x^{(i)},\theta))]$$

